

# UMF — Foundational Logic

*The five-postulate structural framing underlying the Unified Metric Framework*  
Stephen L Browne | Central Coast NSW, May 2026 (AEST) | v1.3

## Preamble

This document records the foundational structural logic of the Unified Metric Framework (UMF) in its current form. It is the layer beneath the formal scaffold and the empirical engagement programme; it states what the framework commits to about the universe at the most basic level, before any specific cosmological observable is engaged.

The framework is not a theory. It is a set of structural commitments about the cosmological lifecycle, the boundary structure of the bounded universe, and the relationship between the universe and its primordial state. The formal mathematical apparatus that would convert these commitments into a theory has not been developed in this document; the present work is preparatory for specialist engagement.

The structural intuitions in this framework — gravitational primacy, a bounded cosmological-scale gravitational event, relaxation as the source of dark-energy phenomenology, finite cosmic lifecycle — have been the author's working position since 1983, formed during a period when both black holes as physical objects and gravitational waves as observables were contested or unconsidered in the rooms the author was in. The formal scaffold is recent; the structural picture is not.

The framework is now publicly available at <https://github.com/OldGreyWhistle/The-UMF> as a citable reference; the current package is V2.3.

Note on revision history: v1.1 (16 May 2026) replaced "dissipation" with "relaxation" throughout in response to cold review feedback. v1.2 (17 May 2026) clarified Postulate 3's commitment on the wavefront's reflective inner boundary, restoring a commitment from the framework lineage (SS-TLGO design document April 2026, Prime-Ordeal blog document January 2026, UMDT and UMGT precursors) that was not explicitly carried into v1.0 of this document. v1.3 (18 May 2026) folds in housekeeping refinements consequent on public GitHub publication: location and timezone consistency (Central Coast NSW, AEST in standard-time months), the GitHub repository reference, and small textual tightening throughout. No structural commitments are changed in v1.3.

## The five postulates

The framework rests on five postulates, stated cleanly below. Each carries a paragraph of explanation about what it commits to and what it does not.

### Postulate 1. A primordial state of gravitational equilibrium exists.

The framework posits a primordial state characterised by gravitational equilibrium. No other assumption is made about this state. The terms "energy," "mass," "time," "space," and "matter" as conventionally defined do not apply to the primordial state; these are categories of the established spacetime regime, not of the regime prior to its establishment. The relationship between the primordial state and the established universe is wholly gravitational. This postulate is a definition of the reference state against which the framework's subsequent commitments are made; it does not

specify the structure or properties of the primordial state beyond the equilibrium condition.

**Postulate 2.** Our universe is created by a gravitational event that establishes a bounded spacetime as a displacement from the primordial equilibrium.

The framework characterises the universe-establishing process by its consequence (a bounded spacetime structure) rather than by intrinsic features of the establishing event. The event is gravitational in nature; it produces a region within which spacetime, mass-energy, and baryonic matter come to exist as a coherent structure. This region is a displacement from the primordial equilibrium — it is what is established when the equilibrium is disturbed gravitationally. The framework takes no position on the mechanism of the establishing process beyond its gravitational character and its result. Whether the establishment was instantaneous, extended in some pre-temporal sense, or had any specific dynamical form is left undetermined.

**Postulate 3.** The established geometry sustains an equilibrium between the primordial state and the bounded spacetime, with the wavefront as a totally reflective inner boundary for established-spacetime content.

Once established, the bounded spacetime maintains structural coherence through the geometric relationships within it and at its boundary. The wavefront — the boundary between the established spacetime and the surrounding primordial state — is an active structural feature with two distinct properties. First, the surface tension at the wavefront's inner surface is at the cosmological surface tension scale ( $P_P \times l_P \approx 7.5 \times 10^{78} \text{ N/m}$ ), which is the maximum sustainable surface tension for the geometric structure. Second, the wavefront's inner surface is totally reflective to all properties of the established spacetime regime — electromagnetic radiation, gravitational waves, matter, particles. The only quantity that crosses the boundary is gravitational tension  $T_G$ , in the direction of relaxation toward primordial equilibrium. This commitment is operational for the framework's wholly-gravitational relationship between Region A and Region C: spacetime content is bounded inside Region A, and the boundary admits only the gravitational displacement that established the regime. Within the bounded region, gravitational tension varies but is bounded by the Planck pressure as the maximum internal stress. The equilibrium is dynamical in the sense that it must be sustained against the natural tendency toward primordial-state relaxation; it is stable in the sense that the established geometry has the structural integrity to maintain itself for an extended cosmic lifecycle.

**Postulate 4.** There is a relaxation effect arising from the tendency of the bounded spacetime to return toward the primordial equilibrium.

The displaced state has a natural tendency to relax back toward the primordial equilibrium. This tendency is realised as a relaxation process operating at the wavefront, where the gravitational displacement is progressively reduced through the wavefront's structural evolution. The framework treats this relaxation as the source of the observed dark-energy phenomenology — the universe's observed acceleration is interpreted as the macroscopic signature of the relaxation, not as the action of an additional energy component within the universe. The specific dynamical form of the relaxation, the rate at which it proceeds, and the detailed mechanism by which the displacement is reduced are open questions that the framework cannot yet resolve. The candidate physical picture is a dissolving polar shell with surface-tension-sustained equilibrium, but the formal apparatus for this picture is open pending

specialist input on quantum gravity programmes applicable at the Planck-pressure regime of the inner surface.

**Postulate 5.** The end state of the cosmic lifecycle matches the beginning state. The conservation budget over the lifecycle replaces the standard concept of energy conservation.

The framework commits to a finite cosmic lifecycle terminating in the return of the bounded spacetime to the primordial equilibrium. The conservation principle that holds across this lifecycle is a gravitational budget — what was displaced is what relaxes back. This is not energy conservation in the standard sense, because energy is a category of the established spacetime regime that does not apply to the primordial state. The relationship between the bounded spacetime and the primordial state is wholly gravitational; the conservation that holds across the boundary is therefore a conservation of the gravitational displacement itself, summed over the lifecycle. The information content of the universe, on this view, transitions rather than is destroyed at the lifecycle's end — but the transition is into a regime where the conventional categories of information do not apply, and the framework takes no position on what the post-transition state contains in any conventional sense.

## **What this foundational logic commits the framework to**

The five postulates above commit the framework to several specific positions that are worth flagging explicitly, because they distinguish the framework from standard cosmology and from other modified-cosmology proposals.

### **A bounded rather than infinite or unbounded universe, with an actively reflective boundary**

The universe in the framework is a bounded region with an active boundary. This contrasts with standard cosmology's typical treatment of the universe as either infinite or as the spatial slice of an extended spacetime manifold without a meaningful boundary at the cosmological scale. The framework's commitment is structural: there is a wavefront, and beyond the wavefront is not more universe but the primordial state to which the universe will eventually return. The wavefront is totally reflective to established-spacetime content, with only the gravitational displacement crossing in the relaxation direction. Spacetime, electromagnetic radiation, gravitational waves, matter, and particles are all bounded inside Region A by the wavefront's reflective inner surface.

### **Gravity as the cosmologically primary phenomenon**

The framework treats gravity as the primary phenomenon establishing the structural conditions within which other physical interactions operate. This is consistent with the empirical foundation laid by Eddington's 1919 demonstration that gravity shapes the path of light and affects the flow of time — gravity is the geometric primary, and electromagnetic and other interactions operate within the geometry that gravity establishes. The framework extends this commitment to the cosmological scale: the universe's structure is gravitationally established and gravitationally maintained, with electromagnetic and matter phenomena as consequences operating within that established geometry. The wholly-gravitational relationship between Region A and Region C through Postulate 3's reflective boundary is the operational form of this gravitational primacy at the cosmological scale.

## Dark-energy phenomenology as boundary relaxation, not field content

The observed acceleration of the universe is, in the framework, the macroscopic signature of relaxation at the wavefront boundary. There is no dark-energy field, no cosmological constant in the sense of an intrinsic energy density of the vacuum, and no quintessence-like component within the universe. The dark-energy phenomenology is what we observe when we measure expansion history of a universe whose gravitational displacement is gradually being reduced through wavefront-boundary relaxation. The framework predicts a monotonic functional dependence of the equation-of-state parameter  $w(z)$  on cosmic time, with the specific functional form determined by the relaxation mechanism (which is open).

## A finite cosmic lifecycle with conservation across the lifecycle

The framework commits to a finite cosmic lifecycle, with the universe being established, sustaining itself for an extended plateau period, entering a relaxation phase, and terminating in return to the primordial equilibrium. The conservation principle that holds is the gravitational budget summed over the entire lifecycle — the displacement created by the establishing event is balanced by the cumulative relaxation across the lifecycle. This replaces the standard treatment in which the universe is assumed to persist indefinitely (or to its heat death without a structurally distinct endpoint), with the framework treating the endpoint as a structurally significant terminal state.

## Time and space as features of the established regime

Time and space, as categories of physics, are features of the established spacetime regime. They do not extend into the primordial state, and questions about what existed "before" the establishing event or what lies "outside" the bounded region are not well-formed in the framework, because the temporal and spatial categories do not apply to the primordial state. The framework treats these as questions that the categories themselves cannot answer, rather than as questions with unknown answers. This is the framework's position on what is sometimes called the "Big Bang singularity" problem and the related question of pre-cosmic time: the questions assume the applicability of categories that the framework restricts to the established regime.

## On the low-entropy initial state

The framework's foundational logic provides an alternative to the standard cosmological treatment of the low-entropy initial state of the universe. The standard treatment treats the early universe as astronomically improbable in entropic terms, with various proposals (Penrose's Weyl curvature hypothesis, eternal inflation with baby universes, anthropic selection from a multiverse) attempting to explain this improbability.

The framework dissolves rather than answers this puzzle. The early universe is, in the framework, a displacement from the primordial gravitational equilibrium. The displacement itself is not low-entropy or high-entropy in any standard sense, because entropy is a thermodynamic concept that requires the spacetime and matter regime to be defined for it to apply, and the primordial state does not admit those categories. The question "why was the early universe so improbable?" is replaced by the question "why did a gravitational displacement occur?" — and gravitational displacements from equilibrium are common in physics, unlike astronomically improbable initial conditions.

This is a legitimate move if the framework can do real work from this foundational position. Whether it can is a question the formal scaffold and the empirical engagement programme address; the foundational logic above only establishes that the framework has a coherent alternative starting point. The framework offers this alternative because the structural intuitions that produced it have been the author's working position since 1983, when standard cosmology's commitments were less settled than they are today and when alternatives to the low-entropy initial-state problem were on the table in mainstream discussion.

## Relationship to the formal scaffold

The foundational logic above is the structural commitment that the formal scaffold of UMF develops. The formal scaffold introduces specific commitments: the Planck force  $F_P$  and Planck pressure  $P_P$  as physical maxima within the bounded spacetime, the gravitational tension field  $T_G$  as the framework's load-bearing dynamical variable, the variable electromagnetic vacuum couplings ( $\epsilon_0$  and  $\mu_0$ ) as environmental properties of the local  $T_G$  value, the gravitoelectric field as the formal apparatus for dark matter phenomenology, the stochastic gravitational wave background internal to Region A as the form of the primordial gravitational signature under the reflective-boundary commitment, and the empirical engagement programme centred on DESI DR3 monotonic  $w(z)$  as the primary near-term test alongside bounded-universe topology constraints from Planck CMB low- $\ell$  data.

The formal scaffold is the operational layer of the framework. The foundational logic is what the formal scaffold develops. A specialist reading the framework should read this foundational logic document first, then the formal scaffold, then the empirical engagement programme. The order matters because each layer rests on the one below it.

A specialist who finds the foundational logic structurally unsound can stop here; there is no point evaluating the formal scaffold if the foundational commitments are rejected. A specialist who finds the foundational logic at least plausible should proceed to the formal scaffold, where the substantive theoretical work begins.